

A32NX RAAS runway awareness callouts

The Honeywell RAAS option of the EGPWS, with the runways of the sim and a generated voice

Branch	pr/18-a32nx-raas
Aircraft	A32NX (A380X OIT flyPad: settings option only)
Commits	b7a7cd707 feat(a32nx/egpws): add the RAAS runway awareness callouts; 22d963d5c fix(a380x/oit): add the RAAS pin program option to the OIT settings pages
Base	upstream master 2baa2b35e (independent)
Status	Voice confirmed in the sim; CI checks pass. Before opening: confirm the licence of the Piper voice model.

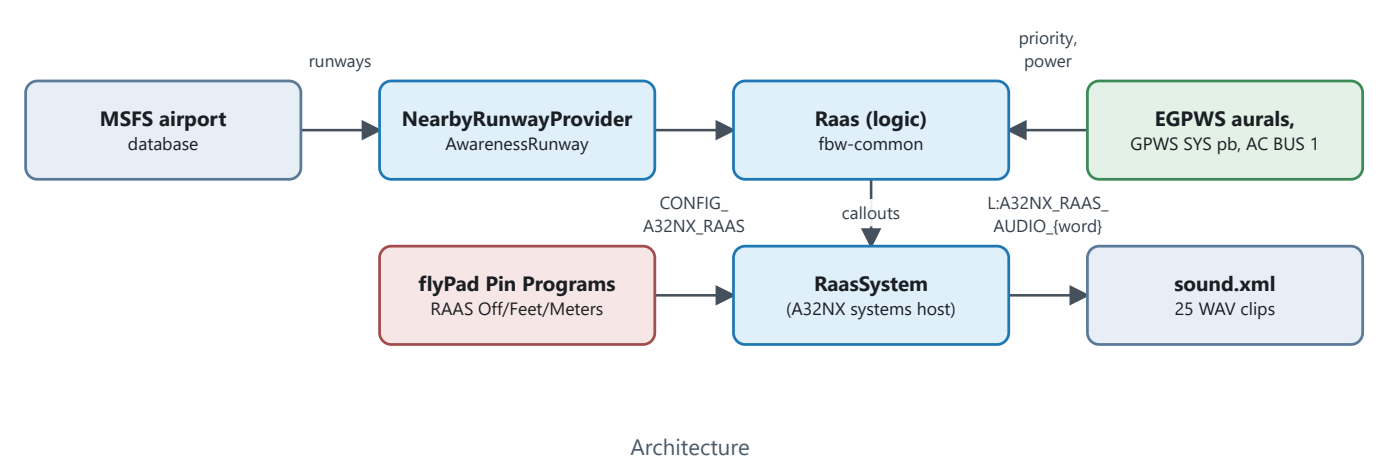
Purpose

The Runway Awareness and Advisory System is an option of the Honeywell EGPWS of the A320 (A320 FCOM GEN, aircraft configuration "RAAS"; used as a line-up aid in PRO-NOR-SOP-11 TAKEOFF RUNWAY CONFIRM). The A32NX did not have it.

Specification (Honeywell RAAS product description 060-4564-001, section 4)

- Approaching runway on the ground and in the air (750 to 300 ft, with the Airbus 450-350 ft suppress window).
- On runway; on taxiway above 40 kt; extended holding on runway.
- Distance remaining at landing and at a rejected takeoff; runway end.
- Approaching short runway and insufficient runway length, with their cautions; taxiway landing.
- The calls yield to the EGPWS aural and run while the EGPWC (AC BUS 1) is powered and the GPWS SYS pushbutton is not OFF.

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
Data, LVar, storage Crew, flypad, external program



Voice

The callouts are played word by word: 25 WAV clips, through the predefined `custom*_sound_NN` event slots of the Wave Data pipeline. The clips are generated by `tools/raas-voice/generate.py` with the Piper text-to-speech engine and its `en_US-ljspeech-medium` voice, trained on the public-domain LJ Speech dataset.

Interfaces

Name	Type	Meaning
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<code>L:A32NX_RAAS_AUDIO_{word}</code>	LVar (Bool)	true while the RAAS says that word (a320-simvars.md, GPWS / TAWS)
<code>CONFIG_A32NX_RAAS</code>	flyPad setting	OFF / FEET (default) / METERS: the unit of the distance callouts

Tests

16 unit tests: `fbw-common/src/systems/shared/src/runway-awareness/Raas.spec.ts`.

Limitations

- Open before the PR: the licence of the Piper voice model `en_US-1jspeech-medium` (its model card); the WAV clips must be redistributable under the repository licence.

PR description (from the PR draft)

From `pr-drafts/18-a32nx-raas.md`, section "Summary of Changes" onwards.

Summary of Changes

The Runway Awareness and Advisory System is an option of the A320's Honeywell EGPWS (A320 FCOM GEN aircraft configuration "RAAS"; used as a line-up aid in PRO-NOR-SOP-11 TAKEOFF RUNWAY CONFIRM). This adds it to the A32NX, implemented from the Honeywell RAAS product description 060-4564-001, section 4:

- **Callouts:** approaching runway on the ground and in the air (750 to 300 ft, with the Airbus 450-350 ft suppress window), on runway, on taxiway above 40 kt, extended holding on runway, distance remaining at landing and at a rejected takeoff, runway end, approaching short runway and insufficient runway length with their cautions, taxiway landing.
- **Runways:** from the sim's own airport database, so no airport map (Navigraph AMDB) is needed.
- **Priorities and power:** the calls yield to the EGPWS aural and run while the EGPWS (AC BUS 1) is powered and the GPWS SYS pushbutton is not OFF.
- **Voice:** played word by word (25 WAV clips) through the predefined `custom_*_sound_NN` event slots of the Wave Data pipeline. The clips are generated by `tools/raas-voice/generate.py` with the Piper text-to-speech engine and its `en_US-ljspeech-medium` voice, trained on the public-domain LJ Speech dataset.
- **Setting:** flyPad Aircraft Options / Pin Programs, RAAS: Off, Feet (default) or Meters (the unit of the distance callouts).
- **OIT:** the A380X OIT flyPad gets the new pin program option too (off, as on the A380X EFB); without it the FBW_TYPECHECK build fails.
- 16 unit tests (`fbw-common/src/systems/shared/src/runway-awareness/Raas.spec.ts`).

Cockpit API Changes

- New LVars `L:A32NX_RAAS_AUDIO_{word}` (Bool, true while the RAAS says that word), documented in `fbw-a32nx/docs/a320-simvars.md` (GPWS / TAWS).
- New flyPad setting `CONFIG_A32NX_RAAS` (OFF / FEET / METERS).

Screenshots (if necessary)

TO ADD: the flyPad Pin Programs RAAS option; a short video or audio of "approaching runway", "on runway" and a distance remaining callout.

References

- Honeywell Runway Awareness and Advisory System (RAAS) product description 060-4564-001, section 4.
- A320 FCOM GEN (aircraft configuration: RAAS), PRO-NOR-SOP-11 (TAKEOFF RUNWAY CONFIRM).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. flyPad Aircraft Options / Pin Programs: RAAS Feet.
2. Taxi towards a runway: "approaching runway ..." at the holding point; line up: "on runway ...".
3. Taxi faster than 40 kt on a taxiway: the taxiway caution.
4. Land: the distance remaining callouts during the rollout (in feet, then with Meters in meters).
5. GPWS SYS pushbutton OFF, or RAAS Off: no callouts.
6. `npx vitest run fbw-common/src/systems/shared/src/runway-awareness/Raas.spec.ts` (16 tests).